



Cloud Computing Fundamentals

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DECEMBER, 2021

NOME: MARCO

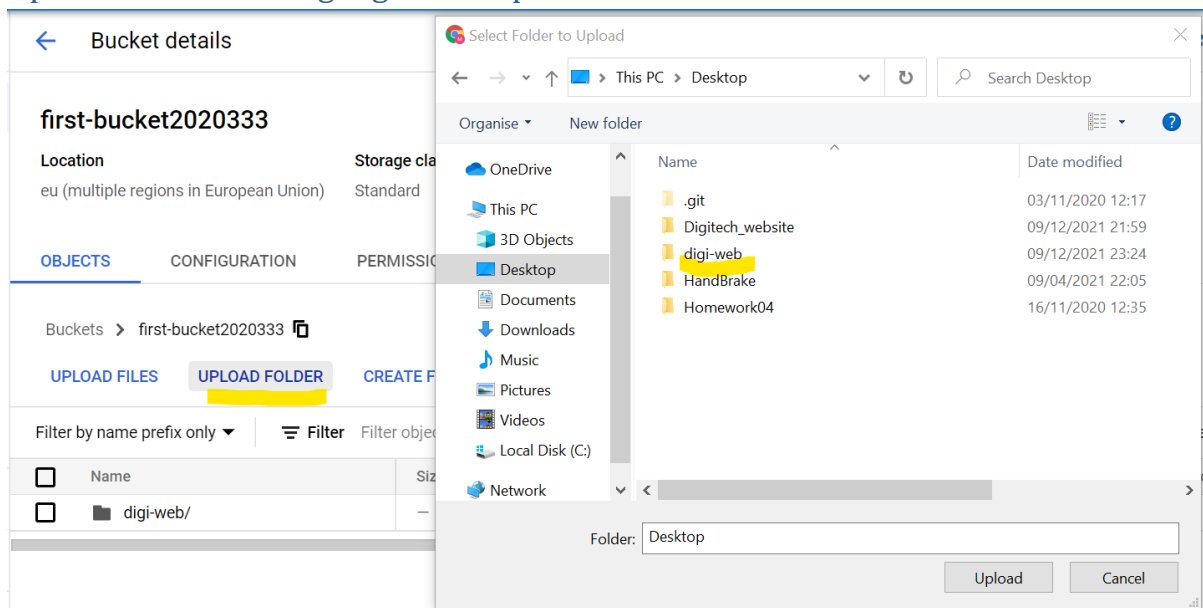
STUDENT NUMBER: 202033

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PART 01

Upload the folder on google cloud plataform



Google SDK Shell – uploading files in my bucket

```
C:\Users\mathe\AppData\Local\Google\Cloud SDK>cd ../../../../
C:\Users>cd C:\Users\mathe\Desktop\Marco
C:\Users\mathe\Desktop\Marco>gsutil cp -r digi-web gs://first-bucket2020333
Copying file://digi-web\.bashrc [Content-Type=application/octet-stream]...
Copying file://digi-web\construction.jpg [Content-Type=image/jpeg]...
Copying file://digi-web\construction2.jpg [Content-Type=image/jpeg]...
Copying file://digi-web\desktop.ini [Content-Type=application/octet-stream]...
\ [4 files][ 29.7 KiB/ 29.7 KiB]
==> NOTE: You are performing a sequence of gsutil operations that may
run significantly faster if you instead use gsutil -m cp ... Please
see the -m section under "gsutil help options" for further information
about when gsutil -m can be advantageous.

Copying file://digi-web\digitech.jpg [Content-Type=image/jpeg]...
Copying file://digi-web\digitech_create.jpg [Content-Type=image/jpeg]...
Copying file://digi-web\index.html [Content-Type=text/html]...
Copying file://digi-web\images\construction.jpg [Content-Type=image/jpeg]...
Copying file://digi-web\images\construction2.jpg [Content-Type=image/jpeg]...
Copying file://digi-web\images\desktop.ini [Content-Type=application/octet-stream]...
Copying file://digi-web\images\digitech.jpg [Content-Type=image/jpeg]...
Copying file://digi-web\images\digitech_create.jpg [Content-Type=image/jpeg]...
| [12 files][ 94.7 KiB/ 94.7 KiB]
Operation completed over 12 objects/94.7 KiB.
```

Linux-web

```
g2020333@linux-web:~$ gsutil cp -r gs://first-bucket2020333/digi-web .
Copying gs://first-bucket2020333/digi-web/.bashrc...
Copying gs://first-bucket2020333/digi-web/bashrc...
Copying gs://first-bucket2020333/digi-web/construction.jpg...
Copying gs://first-bucket2020333/digi-web/construction2.jpg...
/ [4 files][ 32.9 KiB/ 32.9 KiB]
==> NOTE: You are performing a sequence of gsutil operations that may
run significantly faster if you instead use gsutil -m cp ... Please
see the -m section under "gsutil help options" for further information
about when gsutil -m can be advantageous.

Copying gs://first-bucket2020333/digi-web/desktop.ini...
Copying gs://first-bucket2020333/digi-web/digitech.jpg...
Copying gs://first-bucket2020333/digi-web/digitech_create.jpg...
Copying gs://first-bucket2020333/digi-web/images/construction.jpg...
Copying gs://first-bucket2020333/digi-web/images/construction2.jpg...
Copying gs://first-bucket2020333/digi-web/images/desktop.ini...
Copying gs://first-bucket2020333/digi-web/images/digitech.jpg...
Copying gs://first-bucket2020333/digi-web/images/digitech_create.jpg...
Copying gs://first-bucket2020333/digi-web/index.html...
- [13 files][ 98.2 KiB/ 98.2 KiB]
Operation completed over 13 objects/98.2 KiB.
g2020333@linux-web:~$
```

Open the folder

```
g2020333@linux-web:~$ ls -al
total 48
drwxr-xr-x 8 g2020333 g2020333 4096 Dec  9 23:35 .
drwxr-xr-x 3 root     root      4096 Dec  9 20:16 ..
-rw----- 1 g2020333 g2020333  630 Dec  9 22:16 .bash_history
-rw-r--r-- 1 g2020333 g2020333  220 Apr 18  2019 .bash_logout
-rw-r--r-- 1 g2020333 g2020333 3526 Apr 18  2019 .bashrc
drwxr-xr-x 3 g2020333 g2020333 4096 Dec  9 21:27 .config
drwx----- 3 g2020333 g2020333 4096 Dec  9 20:17 .gnupg
drwxr-xr-x 3 g2020333 g2020333 4096 Dec  9 21:27 .gsutil
-rw-r--r-- 1 g2020333 g2020333  807 Apr 18  2019 .profile
drwx----- 2 g2020333 g2020333 4096 Dec 13 15:44 .ssh
drwxr-xr-x 3 g2020333 g2020333 4096 Dec  9 21:27 Digitech_website
drwxr-xr-x 3 g2020333 g2020333 4096 Dec 13 15:47 digi-web
g2020333@linux-web:~$
```

Enabling permissions

```
g2020333@linux-web:~$ sudo chmod 777 /var/www/html
g2020333@linux-web:~$
```

```
g2020333@linux-web:~$ sudo gsutil cp -r gs://first-bucket2020333/digi-web/* . /var/www/html
```

Create a Linux-web

Start your free trial with \$300 in credit. Don't worry - you won't be charged if you run out of credit. [Learn more](#)

DISMISS ACTIVATE

Google Cloud Platform My First Project Search products and resources

Compute Engine VM instances CREATE INSTANCE IMPORT VM OPERATIONS HELP ASSISTANT SHOW INFO PANEL LEARN

Instances INSTANCE SCHEDULE

VM instances are highly configurable virtual machines for running workloads on Google infrastructure. [Learn more](#)

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	linux-web	europe-west2-c			10.154.0.11 (nic0)	35.197.202.211	SSH
☐	linux-web-php	europe-west2-c			10.154.0.10 (nic0)	None	SSH

Related actions DISMISS

Page is working properly external IP address.

VM instances - Compute Engine X Digitech g2020333@linux-web: ~ - Google Chrome

Not secure | 35.197.202.211 ssh.cloud.google.com/projects/awesome-gist-328114/zones/europe-west2-c/instances/linux-web?authuser=0&hl=

see the -m section under "gsutil help options" for further information

about when gsutil -m can be advantageous.

Copying gs://first-bucket2020333/digi-web/desktop.ini...
Copying gs://first-bucket2020333/digi-web/digitech.jpg...
Copying gs://first-bucket2020333/digi-web/digitech_create.jpg...
Copying gs://first-bucket2020333/digi-web/index.html...
Copying gs://first-bucket2020333/digi-web/images/construction.jpg...
Copying gs://first-bucket2020333/digi-web/images/construction2.jpg...
Copying gs://first-bucket2020333/digi-web/images/desktop.ini...
Copying gs://first-bucket2020333/digi-web/images/digitech.jpg...
Copying gs://first-bucket2020333/digi-web/images/digitech_create.jpg...
Copying file:///./bash_logout...
Copying file:///./profile... KiB
Copying file:///./bashrc... 2 KiB
Copying file:///./bash_history...
Copying file:///./config/gcloud/gcs...
Copying file:///./config/gcloud/active_config...
Copying file:///./config/gcloud/configurations/config_default...
Copying file:///./gsutil/credstore2.lock...
Copying file:///./gsutil/goecredcache...
Copying file:///./gsutil/credstore2...
Copying file:///./gsutil/goecredcache.lock...
Copying file:///./Digitech_website/php.html...
Copying file:///./Digitech_website/style.css...
Copying file:///./Digitech_website/images/Perfil-01.png...
Copying file:///./digi-web/desktop.ini...
Copying file:///./digi-web/index.html...
Copying file:///./digi-web/bashrc...
Copying file:///./digi-web/construction2.jpg...
Copying file:///./digi-web/digitech.jpg...
Copying file:///./digi-web/construction.jpg...
Copying file:///./digi-web/index.htm.html...
Copying file:///./digi-web/digitech_create.jpg...
Copying file:///./digi-web/.bashrc...
Copying file:///./digi-web/images/desktop.ini...
Copying file:///./digi-web/images/construction2.jpg...
Copying file:///./digi-web/images/digitech.jpg...
Copying file:///./digi-web/images/construction.jpg...
Copying file:///./digi-web/images/digitech_create.jpg...
- [41 files] [256.2 KiB/256.2 KiB]
Operation completed over 41 objects/256.2 KiB.
g2020333@linux-web:~\$ sudo gsutil cp -r gs://first-bucket2020333/digi-web/* . /var/www/html

Welcome to Digitech!

Bringing Digital Technologies to an Analog world

Student Name: Marco

Student Number: 2020333





Digitech-global-template

Instance temp...

CREATE INSTANCE TEMPLATE

REFRESH

LEARN

SHOW INFO PANEL

Instance templates are saved VM configurations used to create identical VMs, either individually or as part of managed instance groups. [Learn more](#)

Filter

Filter instance templates

☐	Name ↑	Machine type	Image	Disk type	Placement policy ?	In use by	Creation time	Actions
☐	digitech-global-template	e2-medium	debian-10-buster-v20211105	Balanced persistent disk	No policy	digitec... ▼	Dec 13, 2021, 4:26:34 pm UTCZ	⋮

Digitech-europe-group, Digitech-usa-group

Instance templates

+ CREATE VM

CREATE SIMILAR

CREATE INSTANCE GROUP

digitech-global-template

Machine type

e2-medium (2 vCPUs, 4 GB memory)

Reservation

Automatically choose

In use by

[digitech-europe-group](#), [digitech-usa-group](#)

Display device

Turn on a display device if you want to use screen capturing and recording tools.

Turn on display device

Node affinities

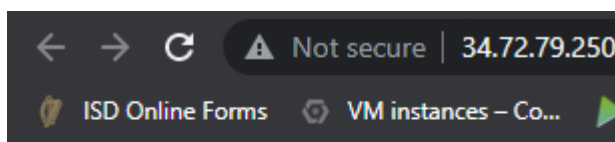
Labels

None

Creation time

13 Dec 2021, 16:26:34

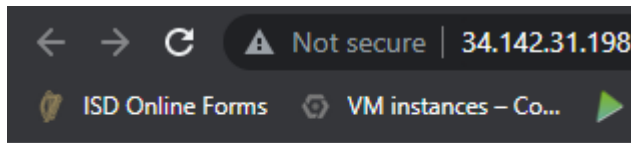
External IP address – Digitech USA



My global website!

digitech-usa-group-n799

External IP address – Digitech Europe



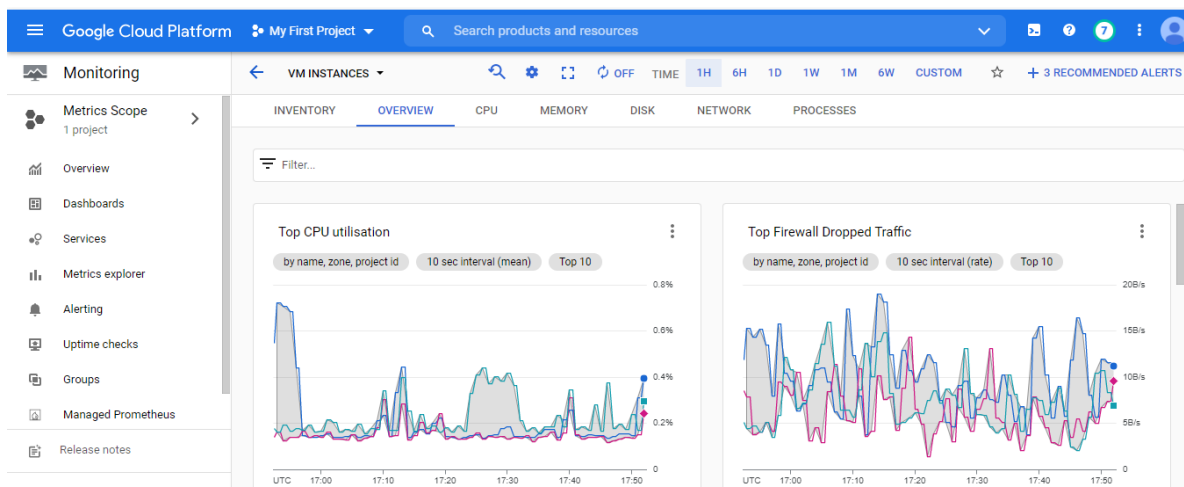
My global website!

digitech-europe-group-jjcq

Before stress Instances

VM instances									
CREATE INSTANCE IMPORT VM OPERATIONS HELP ASSISTANT SHOW INFO PANEL LEARN									
INSTANCES INSTANCE SCHEDULE									
VM instances are highly configurable virtual machines for running workloads on Google infrastructure. Learn more									
Filter Enter property name or value									
☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect	
☐	✓	digitech-europe-group-jjcq	europe-west2-c		digitec...	10.154.0.12 (nic0)	34.142.31.198	SSH	⋮
☐	✓	digitech-usa-group-n799	us-central1-a		digitec...	10.128.0.4 (nic0)	34.72.79.250	SSH	⋮
☐	✓	linux-web	europe-west2-c			10.154.0.11 (nic0)	35.197.202.211	SSH	⋮
☐	⊙	linux-web-php	europe-west2-c			10.154.0.10 (nic0)	None	SSH	⋮

Before stress overview



Install the Linux stress program.

```
g2020333@digitech-europe-group-jjqc:~$ sudo apt install stress
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 5 not upgraded.
Need to get 21.8 kB of archives.
After this operation, 55.3 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian buster/main amd64 stress amd64 1.0.4-4 [21.8 kB]
Fetched 21.8 kB in 0s (1312 kB/s)
Selecting previously unselected package stress.
(Reading database ... 49106 files and directories currently installed.)
Preparing to unpack .../stress_1.0.4-4_amd64.deb ...
Unpacking stress (1.0.4-4) ...
Setting up stress (1.0.4-4) ...
Processing triggers for man-db (2.8.5-2) ...
g2020333@digitech-europe-group-jjqc:~$ uptime
 18:22:37 up  1:50,  1 user,  load average: 0.09, 0.04, 0.00
g2020333@digitech-europe-group-jjqc:~$
```

Stress successful

```
g2020333@digitech-europe-group-jjqc:~$ sudo stress --cpu 12 --timeout 100
stress: info: [9472] dispatching hogs: 12 cpu, 0 io, 0 vm, 0 hdd
stress: info: [9472] successful run completed in 100s
g2020333@digitech-europe-group-jjqc:~$
```

New Virtual machines created with stress

Start your free trial with \$300 in credit. Don't worry - you won't be charged if you run out of credit. [Learn more](#)

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Compute Engine

Virtual machines

VM instances

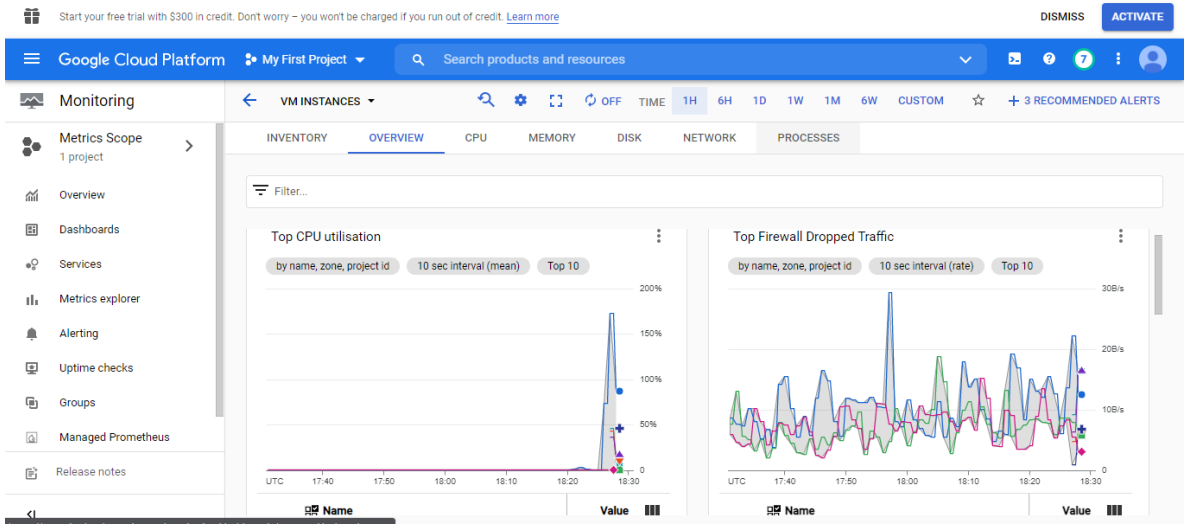
CREATE INSTANCEIMPORT VMOPERATIONSHELP ASSISTANTSHOW INFO PANELLEARN

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	digitech-europe-group-3hvm	europa-west2-c		digitec...	10.154.0.16 (nic0)	34.89.11.38	SSH
☒	digitech-europe-group-9p3w	europa-west2-c		digitec...	10.154.0.15 (nic0)	35.242.151.248	SSH
☒	digitech-europe-group-9vrz	europa-west2-c		digitec...	10.154.0.14 (nic0)	35.189.125.251	SSH
☒	digitech-europe-group-jjqc	europa-west2-c		digitec...	10.154.0.12 (nic0)	34.142.31.198	SSH
☒	digitech-europe-group-r97g	europa-west2-c		digitec...	10.154.0.13 (nic0)	35.230.132.237	SSH
☒	digitech-usa-group-	us-central1-a		digitec...	10.128.0.4 (nic0)	34.72.79.250	SSH

<https://console.cloud.google.com/compute/instanceTemplates/list?project=awsco-299>

The rise and fall of CPU utilization



Static IP Address:

← Load balancer details

EDITDELETE

Frontend

Protocol ↑	IP:Port	Certificate	SSL Policy	Network Tier ?
HTTP	34.117.111.85:80	-		Premium

Host and path rules

Hosts ↑	Paths	Backend
All unmatched (default)	All unmatched (default)	backend-marco

Backend

Backend services

1. backend-marco

Endpoint protocol	Named port	Timeout	Health check	Cloud CDN
HTTP	http	30 seconds	healthcheck-marco	Disabled

✓ ADVANCED CONFIGURATIONS

PART 02 – DIGIZILLA

Create a Linux virtual Machine

VM instances									
CREATE INSTANCE									
OPERATIONS									
HELP ASSISTANT									
LEARN									
SHOW INFO PANEL									
INSTANCES									
INSTANCE SCHEDULE									
VM instances are highly configurable virtual machines for running workloads on Google infrastructure. Learn more									
Filter Enter property name or value									
☐	Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect	
☐	✓	digitech-europe-group-9vrz	europe-west2-c		digitec...	10.154.0.14 (nic0)	35.189.125.251	SSH	
☐	✓	digitech-usa-group-n799	us-central1-a		digitec...	10.128.0.4 (nic0)	34.72.79.250	SSH	
☐	○	digizilla	europe-west2-c			10.154.0.17 (nic0)	None	SSH	

Upload DigiTech website to Linux using FileZilla

Public Key stored

My First Project

metadata

Metadata

Metadata

SSH Keys

Edit

All instances in this project inherit these SSH keys. [Learn more](#)

Username

Key

rsa-key-20211213

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAjT/lenbezSjLRN32zWDMmZzhF3mNsYes7/G2BJBpt7Lbrfl uMYAUoEx7/ITo1NdNDp3Oyy774pBQnt5m9Z35ZVShOROmCDeTu9gaVnAXkTTHA+LygEDoZUpN +Qht7Y2eZJrqUFEQXCTXdF2eBuM

Equivalent REST

Putty Key Generator

File Key Conversions Help

Key

Public key for pasting into OpenSSH authorized_keys file:

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAjT/lenbezSjLRN32zWDMmZzhF3mNsYes7/G2BJBpt7Lbrfl uMYAUoEx7/ITo1NdNDp3Oyy774pBQnt5m9Z35ZVShOROmCDeTu9gaVnAXkTTHA+LygEDoZUpN +Qht7Y2eZJrqUFEQXCTXdF2eBuM

Key fingerprint: ssh-rsa 2048 SHA256:pfCnNwUFFd7+zly9YORictcADPWk460WWTzKnt7P1D0

Key comment: rsa-key-20211213

Key passphrase:

Confirm passphrase:

Actions

Generate a public/private key pair

Load an existing private key file

Save the generated key

Generate

Load

Save public key

Save private key

Parameters

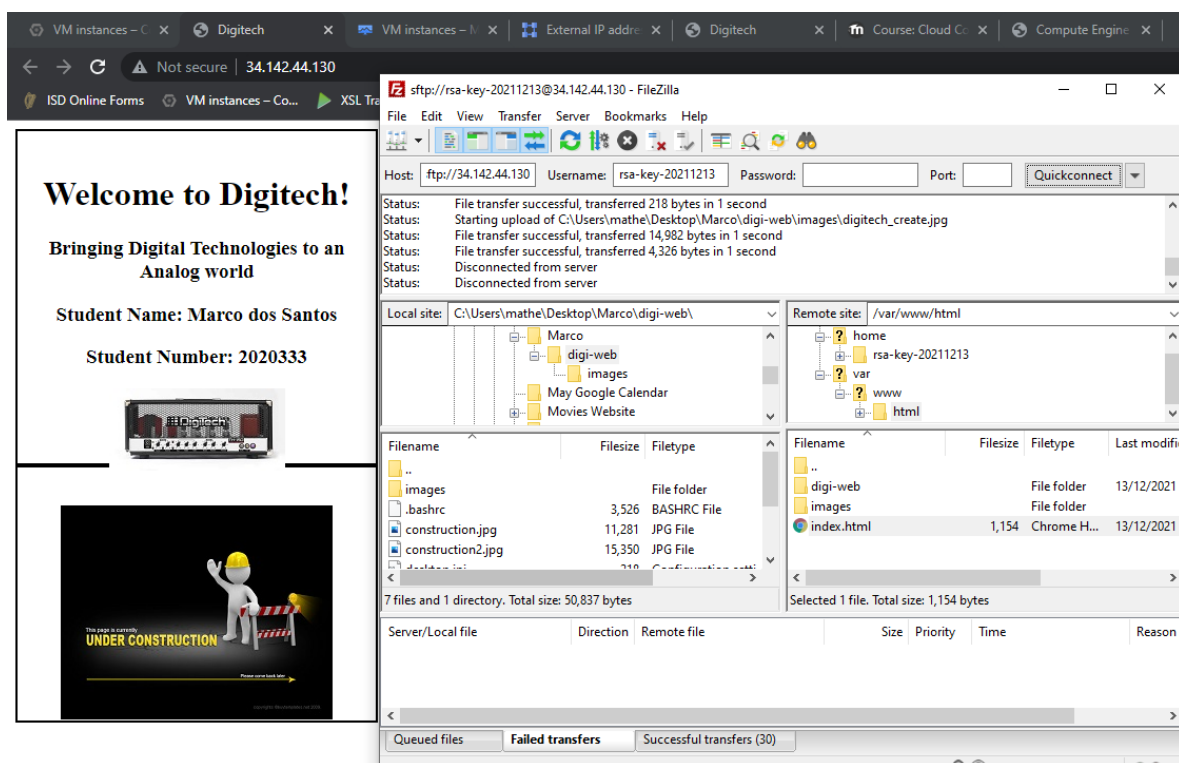
Type of key to generate: ☒ RSA ☐ DSA ☐ ECDSA ☐ EdDSA ☐ SSH-1 (RSA)

Number of bits in a generated key: 2048

Changing permissions to open my website:

```
Processing triggers for libc-bin (2.28-10) ...
g2020333@digizilla:~$ stat -c "%a %n" /var/www/html
755 /var/www/html
g2020333@digizilla:~$ sudo chmod 777 /var/www/html
g2020333@digizilla:~$ stat -c "%a %n" /var/www/html/index.html
644 /var/www/html/index.html
g2020333@digizilla:~$ sudo chmod 777 /var/www/html/index.html
g2020333@digizilla:~$
```

DigiZilla to display the web page:



CHALLENGE

Installing gcsfuse

```
g2020333@linux-vm-with-storage: ~ - Google Chrome
ssh.cloud.google.com/projects/awesome-gist-328114/zones/europe-west2-c/instances/linux-vm-with-storage?authuser=0&hl=en_GB&projec

g2020333@linux-vm-with-storage:~$ export GCSFUSE_REPO=gcsfuse-`lsb_release -c -s`
g2020333@linux-vm-with-storage:~$ echo "deb http://packages.cloud.google.com/apt $GCSFUSE_REPO main" | sudo tee /etc/apt/sources.list.d/gcsfuse.list
deb http://packages.cloud.google.com/apt gcsfuse-buster main
g2020333@linux-vm-with-storage:~$ curl https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo apt-key add -
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  2537  100  2537    0     0  103k      0 --:--:-- --:--:-- --:--:-- 103k
OK
g2020333@linux-vm-with-storage:~$ sudo apt update
Hit:1 http://deb.debian.org/debian buster InRelease
Hit:2 http://deb.debian.org/debian buster-updates InRelease
Hit:3 http://deb.debian.org/debian buster-backports InRelease
Hit:4 http://security.debian.org/debian-security buster/updates InRelease
Hit:5 http://packages.cloud.google.com/apt cloud-sdk-buster InRelease
Hit:6 http://packages.cloud.google.com/apt google-cloud-packages-archive-keyring-buster InRelease
Get:7 http://packages.cloud.google.com/apt gcsfuse-buster InRelease [5388 B]
Hit:8 http://packages.cloud.google.com/apt google-compute-engine-buster-stable InRelease
Fetched 5388 B in 1s (5994 B/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
5 packages can be upgraded. Run 'apt list --upgradable' to see them.
g2020333@linux-vm-with-storage:~$ sudo apt install gcsfuse
Reading package lists... Done
Building dependency tree
Reading state information... Done
gcsfuse is already the newest version (0.38.0).
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
g2020333@linux-vm-with-storage:~$
```

Challenge complete!

```
g2020333@linux-vm-with-storage:~$ gcsfuse --only-dir digi-web first-bucket2020333 ~/marco
2021/12/13 19:19:57.284781 Start gcsfuse/0.38.0 (Go version go1.17.3) for app "" using mount point: /home/g2020333/marco
2021/12/13 19:19:57.302234 Opening GCS connection...
2021/12/13 19:19:57.486530 Mounting file system "first-bucket2020333"...
2021/12/13 19:19:57.493290 File system has been successfully mounted.
g2020333@linux-vm-with-storage:~$ cd marco
g2020333@linux-vm-with-storage:~/marco$ ls
bashrc  construction.jpg  construction2.jpg  desktop.ini  digitech.jpg  digitech_create.jpg  index.html
g2020333@linux-vm-with-storage:~/marco$
```

Reference:

Video: mount google cloud storage in ubuntu[ACCESS:11/12/21]:
Available:<https://www.youtube.com/watch?v=PCqL38coas8>